

Honing USD: Lessons Learned and Workflow Enhancements at Walt Disney Animation Studios

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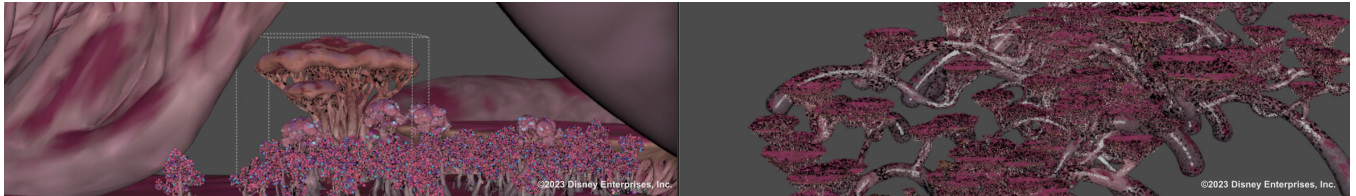


Figure 1: *Strange World*'s lung platform set, rendered in Hydra for usdview. After composition optimizations, stage open performance times saw a 47.4 percent speedup, and with scenegraph instancing, a 56.2 percent speedup. 430,000 prims, 183,000 meshes. ©2023 Disney.

ABSTRACT

During development for *Strange World*, we expanded on an existing Pixar's Universal Scene Description (USD) based pipeline, transitioning Disney's XGen to be authored in USD and paving the way for more departments to explore Houdini-based workflows. Due to the increase in data on our USD stages, in combination with remaining optimization opportunities from the first iteration of our USD pipeline, Walt Disney Animation Studios presents techniques and lessons learned to optimize our asset structure and redesign our change management pipeline to uphold interactive performance, all while sustaining existing artist workflows.

CCS CONCEPTS

• General and reference → Design; • Computing methodologies → Animation.

KEYWORDS

pipeline, USD, animation, production

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1 INTRODUCTION

Pixar's Universal Scene Description (USD) integration efforts can often be conceptualized with the principle that USD is primarily an interchange format [Pixar 2016]. For such efforts, platform integration could be scoped to only manage translation and consumption of USD. The *Encanto* pipeline established a foundational USD asset and shot structure [Miller et al. 2022], with USD as our primary authoring format. *Strange World* technical teams continued that work by transitioning Disney's XGen to be backed by USD. The ambitious environments of the film prompted exploring improvements to our existing procedural systems and marrying them to the existing core USD pipeline. Furthermore, transitioning Disney's XGen to USD allowed more artist departments to preview and powerfully author the distinct cell-biology inspired vegetation within Houdini.

Disney's XGen was built as a flexible layering system for our hair and vegetation generation. Artists could share procedural recipes and add sparse edits across different asset variants. These workflows were intuitive to fit to USD, and Disney's XGen could maintain its own in-memory representation that syncs to USD. However, other workflows that required destructive edits exposed limitations in our USD asset structure. Asset variants shared data through references, disallowing destructive edits after composition in USD. These edits couldn't be reflected in the composed USD state, desyncing it from Disney's XGen. The transition effort also occurred during asset production, prohibiting a disruptive asset restructure solution. Thus, we used additional references at different prim anchors for procedural recipes that diverged from upstream asset variants' recipes. In the worst case, Disney's XGen to USD transition increased the number of prims by the tens of thousands in a production shot. This additional data in conjunction with additional optimization opportunities seen on *Encanto*'s pipeline motivated us to improve our interactive performance of our USD data and the pipeline's change management.

2 ON-DEMAND SHOT WORKFLOWS

With Disney’s first USD pipeline, we used payloads by default for all arcs, deferring composition within the payload in unloaded prims. We used payloads at the asset root for assemblies, and at multiple lower anchors for geometry. However, tools needed to load the asset root to access asset metadata identifying asset variants, names, and types, which then forced the loading of lower geometry payloads. Our inability to stage payload unfurling negated the intent to have lighter-weight on-demand shot workflows or scene level queries. As a solution, we updated our shot to use references at the asset root, with a single payload at the geometry subtree root. To make the transition, every pipeline tool that relied on payload manipulation had to be updated to use references to maintain predictable opinion strength during shot production. Post-transition, scene level queries could run on unloaded shots, and since Houdini provided stage load and unload tools, Houdini artists could immediately use unloaded stages, previewing only bounding boxes, and resolving only the assets they were working on. Baked animation on parent transforms were also now referenced, providing accurate animated bounding boxes in an unloaded stage. Further, change management within USD’s engine did not have to resolve prims within each component subtree, increasing authoring performance.

3 ASSET COMPOSITION OPTIMIZATION

Before *Strange World*, our assets contained multiple payloads, as well as inherits arcs to push opinions of higher strength during modeling workflows. inherits arcs are powerful and can pick up changes across different arcs, whereas other arcs can’t introduce new opinions once they are composed. However, inherits arcs are more expensive, as the composition engine must check for opinions across potentially many sites and layers. Many non-asset-root inherits arcs we introduced into our assets added compounding overhead in our shot workflows. Because we didn’t author opinions through those inherits arcs after modeling, flattening our composition arcs on model publish to a single payload optimized all assets without interrupting any workflows. The payload anchor prim could also be tagged as instanceable to enable USD’s implicit scenegraph instancing.

4 NOTICE CONSUMPTION

After stage writes, USD emits notices that can be registered by consumers (e.g. real-time visualization). Consumers invalidate the in-memory state for each prim path sent in a USD notice; consumer resyncs were expensive if the paths seen in notices were either too granular or broad. For example, with sublayer changes, USD sends a notice with the root prim path, warning consumers that the entire scene’s state is invalidated. For some workflows, we were able to make assumptions about which specific asset paths required a resync, no longer necessitating full scene resync within a consumer. For other actions, like saving (which, in our pipeline, required layer muting and unmuting), we could bypass resyncing entirely. We create custom USD pre- and post- layer manipulation notices to send out to consumers. For a pre-opn notice, consumers would temporarily unregister their native USD notice listener, then

filter the prim paths based on workflow assumptions. For a post-operation notice, consumers resync only the paths of interest, and reinstate the native USD notice listener.

5 SCENEGRAPH INSTANCING

Scenegraph instancing could be used as a way to reduce the compositional weight of assets that are referenced multiple times in sets as *hero* assets (e.g. copies of assets, instead of demoted to instances). Scenegraph instancing was attractive because of its implicit nature; environments artists could set-dress without worrying about optimization. Workflows could use flexible promotion to non-instanced and demotion to instanced state per asset copy. Modifications would be added directly on a promoted asset, rather than pushed into a large list of parameters per instance like with point instancer workflows, leading to easier data inspectability and management. Lastly, scenegraph instancing implicitly provides large subtree deduplication, even with very few copies, and reduce the cost of particularly large assets with few copies. Scenegraph instancing was fully supported by converting all component data to be encapsulated within instanceable payloads on publish to shots, maintaining opinions introduced as local overrides in working scenes. We also updated all tooling to handle the changes in scene graph traversal.

6 RESULTS AND FUTURE WORK

Table 1: Scene load performance measurements with the various optimizations discussed, cumulatively applied. Measured on a set with 192 assets (651,545 prims).

	Total	USD Composition	GPU Visualization
No optimization	90.2s	51.1s	3.2s
Asset flattened	53.0s	1.6s	28.0s
Assets flattened, instanced	36.2s	1.7s	17.4s
Assets flattened, instanced, and custom notices	8.1s	0.5s	0.1s

Scenegraph instancing will be expanded upon as an authoring workflow, where we will use edit context management in USD to broadcast changes to multiple instances without breaking instancing. We also are improving the asset variant representation in USD to avoid the reference encapsulation encountered in our Disney’s XGen to USD transition. Variants would share data as sublayers instead of references, and allow us to properly clean up data on the layers that introduced them. Further, development of more Houdini based workflows allow us to utilize Houdini’s multi-stage workflows, and will allow us to move away from layer muting and unmuteing, which still require large recomposition, even with improved change management.

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